KCE PATHWAYS · GRADES 4-5 · UNIT 3: BUILDING THINGS · WEEK 11 · SCIENCE

Matter and Mixtures

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. What is matter? Name three things that are matter and one thing that is not.

2. Predict: When a log burns down to a small pile of ash, does most of its matter disappear? Where do you think it goes?

3. Draw a mixture you know (a salad, trail mix, a toy box). Label two parts that keep their own properties.

4. Predict: A cup of water is weighed. Then a spoonful of salt is stirred in until it dissolves. Will the cup weigh more, less or the same? Why?

5. What is one thing you wonder about mixing or changing materials?

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Show What I Learned

Use evidence from today's lesson to explain your thinking.

1. Explain the difference between a mixture and a solution. Give one example of each.

2. Use the class data: why did Trial 3 lose mass but Trial 4 did not? Use the word 'gas' in your answer.

3. Write the steps you would use to separate a mixture of sand, salt and iron filings. Name the property each step uses.

4. Choose Alice Ball or Mario Molina. How did knowing the properties of matter help that scientist help people?

★ **Challenge:** With a parent, put a little baking soda in a bottle and a little vinegar in a balloon. Stretch the balloon over the bottle's top, weigh it, then tip the vinegar in. Weigh it again. Did the mass change? Explain why.

Looking Ahead: Next week we explore energy and electricity: forms of energy, circuits, and why some materials let electricity flow while others block it. Question to bring: Why are electrical wires made of metal on the inside and plastic on the outside?